

IVAN ARIAS

GIS Analyst · Geospatial Data & Web GIS

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Languages: Spanish (native) · English (C1) · Dutch (A1) · Work authorisation: [EU]

PROFILE

GIS and spatial data specialist with 13 years of professional geospatial work in the energy sector. At PDVSA (Venezuela and Ecuador, 2005–2018) built and maintained field-development geodatabases and geologic map series in ArcGIS, integrating well-log, core and seismic data across multiple basins; MSc research applied geostatistical spatial modelling across 65 exploration wells. Since 2022, extended that foundation into Python, PostGIS, spatial SQL, FastAPI and Leaflet, building full-stack web GIS applications.

CORE SKILLS

Geospatial: ArcGIS · QGIS · PostGIS · spatial SQL · GeoJSON · Leaflet · geodatabase design and maintenance · thematic mapping · cartography and map-series production · coordinate reference systems · geostatistics and 3D spatial modelling

Data & backend: Python · SQL · PostgreSQL · FastAPI · Flask · SQLAlchemy · Alembic · REST APIs · Plotly Dash

Web: JavaScript · React · Astro · Vite · Tailwind CSS

Platform: Git · Linux · Docker · AWS · Terraform · CI/CD

Subsurface (2005–2018): Petrel · OpenWorks · Landmark · StratWorks · IRAP RMS · WellCad · sequence stratigraphy · reservoir characterisation · well-log analysis · petrography

SELECTED PROJECTS

GIS Competencies — Evidence-based portfolio documenting core GIS competencies: workflows, SQL and spatial data QA.

Oil & Gas GIS Web App — Full-stack web GIS for field infrastructure (boreholes, pipelines, licence boundaries). Leaflet and vanilla JavaScript frontend; FastAPI, PostgreSQL and PostGIS backend with viewport-based spatial loading, bounding-box queries, GeoJSON responses and zoom-dependent layer rendering. Deployed on Render. Demo: maps.hcoco1.com

Leaflet Vanilla JS Web Map — Framework-free spatial web application: layer management, geospatial interaction patterns and responsive map UI. Demo: gis.hcoco1.com

Student Performance Dashboard — Python and Plotly Dash analytics dashboard for academic performance data. Demo: dashboard.hcoco1.com

Full portfolio (20+ projects, including cloud and security coursework): hcoco1.com/portfolio

EXPERIENCE

GIS & Subsurface Spatial Data · PDVSA · Venezuela & Ecuador · 2005–2018 (13 years)

Laboratory → drilling → exploration: Sedimentologist, Geological Laboratory (2005–2009) · Operational Geologist, Drilling (2009–2011) · Sedimentologist, Exploration (2011–2013) · Stratigrapher, PDVSA Ecuador rotation (2013–2015) · Sedimentologist & Stratigrapher, Exploration (2015–2018).

- Produced thematic maps of oil exploration fields in ArcGIS to support early-stage exploration screening and evaluation, and built and maintained the underlying field-development geodatabases used by cross-functional exploration and drilling teams.
- Constructed regional stratigraphic frameworks and interpreted depositional systems across the Barinas-Apure Basin, Santa Elena Fold Belt and Progreso Basin.
- Integrated well-log, core and seismic data for subsurface correlation, spatial modelling and prospect evaluation supporting exploration risk assessment.
- Delivered laboratory characterisation (core description, petrography, XRD/XRF) for reservoir-quality evaluation, and real-time wellsite interpretation during active drilling.

Independent Geoscience Consultant · Venezuela · 2018–2019 — Sedimentology, stratigraphy and subsurface interpretation on a consulting basis (ArcGIS, Petrel).

Additional experience · 2019–present — Full-time employment maintained throughout the career transition: Wasco, Den Haag (2026–) · Fidelity Fulfilment and CEVA Logistics, 's-Hertogenbosch (2025–26) · Amazon, Orlando FL (2020–25), 99%+ accuracy at 2,000+ parcels/day · self-employed landscaping, Orlando FL (2019–20).

EDUCATION & TRAINING

Geological Engineer · Central University of Venezuela · 1999–2005

MSc Geological Sciences — coursework and thesis research completed · Central University of Venezuela · 2009–2013. Thesis: geostatistical static characterisation of Barinas Basin reservoirs — spatial modelling across 65 exploration wells (Petrel, ArcGIS, Landmark, IRAP RMS).

Full-Stack Engineering · Flatiron School · 2023 — Python, Flask, React, SQL, SQLAlchemy

Cloud Mastery · Digital Cloud Training · 2024 — AWS, Terraform, Docker, Kubernetes

Cybersecurity · Fullstack Academy · 2024 — Python, Linux, red team / blue team